

Operational-Duration Boundaries: An Adversarial Test of Physical Transition, Statistical Crossover, and Mathematical Artefact

Matěj Rada

MatejRada@email.cz

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Abstract

A precise fitted crossing is not by itself a physical boundary. We attack four mechanisms giving an apparent boundary near 30 ms: a narrow shared transition, a broad smooth crossover, an acquisition gate, and a heterogeneous mixture. Three positive-affine clock charts are calibrated to one operational-duration object. A deterministic synthetic battery, seed 20260920, applies bootstrap uncertainty, threshold-width separation, held-out logistic versus monotone prediction, and clock quantization from 0.05 to 5 ms. All four mechanisms return reproducible midpoints. Hence midpoint reproducibility, cross-clock agreement, and small uncertainty are necessary but not sufficient for physicality. What survives is a conditional theorem requiring cross-clock closure, resolution convergence, model adequacy, physical-versus-pipeline interventions, and exclusion of hidden mixtures. No universal 30 ms constant is established.

1 Operational question

Let $Y \in \{0, 1\}$ record distinguishability at operational duration T and let $p(T) = \Pr(Y = 1 \mid T)$. For a declared level p_* ,

$$T_{p_*} = \inf\{T : p(T) \geq p_*\}.$$

Every monotone crossover can possess such a number. Existence and precise estimation of T_{p_*} therefore do not establish a physical boundary.

Definition 1 (Boundary packet). *For $p_- < p_* < p_+$ define*

$$\mathcal{B} = (T_{p_*}, I_{p_*}, W_{p_-, p_+}), \quad W_{p_-, p_+} = T_{p_+} - T_{p_-}.$$

Here I_{p_} is location uncertainty and W is intrinsic transition width. These are logically and empirically distinct.*

2 Adversarial battery

The synthetic experiment uses 44 durations from 12 to 55 ms, 160 Bernoulli trials per duration and clock, three affine clock charts, two calibration anchors, and 80 bootstrap replicates per chart and scenario. The attacks are: affine-chart closure; bootstrap location uncertainty; 10–90% width and 30–70% threshold sensitivity; quantization at 0.05, 0.25, 1, 2, and 5 ms; held-out logistic versus isotonic prediction; acquisition-gate injection; and latent-mixture injection.

Table 1: Executed synthetic battery. Values are milliseconds.

Mechanism	midpoint	clock SD	mean 95% CI	$W_{10,90}$	quant. shift
sharp shared	30.09	0.08	0.42	7.67	0.11
smooth crossover	29.69	0.19	1.03	47.99	0.09
instrument gate	26.86	0.10	0.82	25.60	0.08
heterogeneous mixture	30.44	0.19	0.63	19.35	0.04

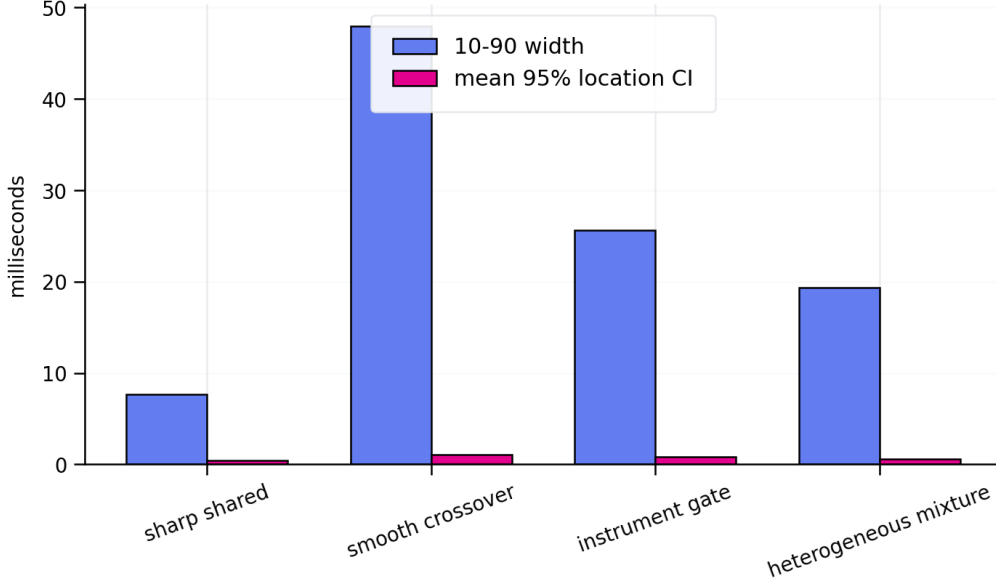


Figure 1: Location uncertainty versus intrinsic width. A precise midpoint can summarize a broad process.

3 Executed results

The narrow shared witness is recovered near 30 ms with small cross-clock dispersion and narrow width. The smooth crossover also yields a stable center but a much wider transition. The instrument-gate witness creates an apparent boundary in the observation pipeline. The pooled mixture creates one midpoint although neither constituent has that midpoint as its individual transition.

Proposition 1 (Midpoint non-identifiability). *Cross-clock agreement of $T_{1/2}$, small sampling uncertainty, and stability under quantization do not jointly identify a physical boundary.*

Proof. The executed witnesses have incompatible generating mechanisms yet all admit reproducible midpoints. One is smooth, one is pipeline-gated, and one is a pooled mixture. Therefore the listed observables are not injective with respect to physical mechanism. $\square$

Theorem 1 (Conditional operational physicality). *Assume at least three independent clocks establish a held-out cocycle-consistent affine duration object. Let a frozen boundary packet reproduce across charts, devices, sessions, saturated feature families, and declared physical interventions. Assume its location and width converge under increasing resolution; residuals have no declared clock, pipeline, environment, or subgroup dependence; acquisition gates and latent mixtures are excluded within the uncertainty model; and smooth competitors predict worse on held-out data. Then the packet is supported as a physical transition of the tested operational system on the tested domain.*

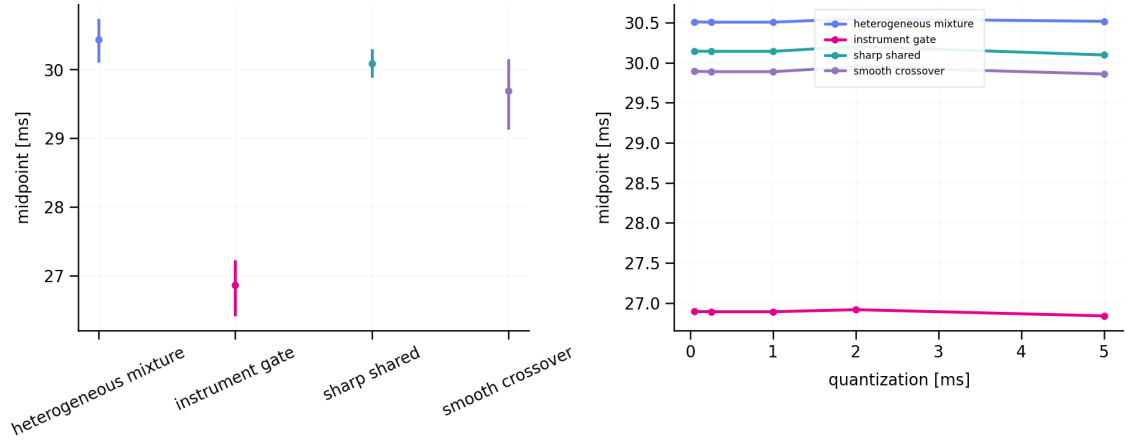


Figure 2: Cross-clock midpoint intervals and quantization response. Stability is necessary, not identifying.

Proof. Affine closure removes chart origin and unit dependence up to external scale. Resolution convergence rejects a bin-lattice crossing. Cross-device and crossed physical/pipeline interventions reject a device-local gate. Residual and subgroup gates reject declared nuisance and pooling explanations. Held-out comparison rejects declared smooth alternatives. The conclusion remains conditional because undeclared observationally equivalent mechanisms may exist. $\square$

Corollary 1 (Status of 30 ms). *The present battery establishes a falsification architecture, not a universal 30 ms constant. A physical wall-tap experiment must estimate the complete boundary packet and close the theorem’s gates.*

4 Locked ladder

B0: ordered records. B1: three-clock affine and cocycle closure. B2: saturated observable state and covariance. B3: frozen held-out distinguishability law. B4: boundary packet. B5: chart, threshold, model, and resolution attacks. B6: pipeline, intervention, and subgroup exclusion. B7: conditional physical transition on the tested domain.

5 Non-claims

The battery is synthetic. It does not re-estimate the earlier 35.58 ms acoustic-overlap witness, establish a perceptual or neural threshold, prove a non-analytic singularity, identify unique microscopic mechanism, or convert the remembered 30 ms neighborhood into a fundamental temporal quantum.

6 Reproducibility

Seed 20260920. The attached ledger contains per-clock midpoint estimates, bootstrap intervals, transition widths, resolution attacks, and model scores. The Python source, JSON record, CSV ledgers, and figures are bundled with this paper. Numerical results are outputs, not theorem premises.

7 Conclusion

No curve-only statistic decides physicality. Operational duration makes cross-clock reconstruction possible, while the boundary packet prevents location precision from being confused with sharpness. The decisive experiment is crossed intervention: alter the physical overlap mechanism while holding the pipeline fixed, then alter the pipeline while holding the physical mechanism fixed. What survives is a conditional, falsifiable boundary claim, not “30 ms is fundamental.”